Moral Hazard - Part I

EC 387 – Health Economics

Professor Pauline Mourot

Back to context

Grossman model (1972) of demand for *health*

Health:

- Enters utility directly
- Is an investment
- Is an input in production of health and of productive time

What about the demand for ~~health~~ *healthcare*?

Some definitions

Moral hazard:

Moral hazard is the change in an individual's behavior that occurs after entering into a contract because the individual does not bear the full consequences of their actions, typically due to imperfect observability or enforcement.

Two types of moral hazard:

- **Ex-ante moral hazard** (Ehrlich and Becker, 1972): change in behavior *before* an event occurs (e.g., risk-taking)
- **Ex-post moral hazard** (Pauly, 1968): change in behavior *after* an event occurs (e.g., consumption of care)

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Examples of moral hazard

- Banks engaging in excessive risk-taking knowing that the government will bail them out
- Individuals driving recklessly when they have car insurance
- More generous unemployment insurance may cause some workers to remain out of work longer
- Academic researchers slacking off after tenure
- Individuals consuming more healthcare when they have health insurance

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Incentives in research? *Hill & Stein (2025)*

RACE TO THE BOTTOM: COMPETITION AND QUALITY IN SCIENCE*

RYAN HILL AND CAROLYN STEIN

This article investigates how competition to publish first and thereby establish priority affects the quality of scientific research. We begin by developing a model where scientists decide whether and how long to work on a given project. When deciding how long they should let their projects mature, scientists trade off the marginal benefit of higher-quality research against the marginal risk of being preempted. Projects with the highest scientific potential are the most competitive because they induce the most entry. Therefore, the model predicts these projects are also the most rushed and lowest quality. We test the predictions of this model in the field of structural biology using data from the Protein Data Bank (PDB), a repository for structures of large macromolecules. An important feature of the PDB is that it assigns objective measures of scientific quality to each structure.

Hill, Ryan, and Carolyn Stein. *"Race to the bottom: Competition and quality in science."* The Quarterly Journal of Economics 140, no. 2 (2025): 1111-1185.

Economists after tenure

Journal of Economic Perspectives—Volume 32, Number 1—Winter 2018—Pages 179–194

Do Economists Swing for the Fences after Tenure?

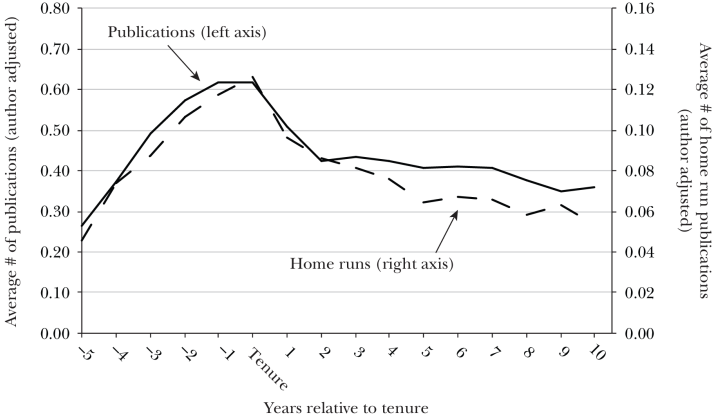
Jonathan Brogaard, Joseph Engelberg, and Edward Van Wesep

Source: https://www.aeaweb.org/articles?id=10.1257/jep.32.1.179

Economists after tenure

Figure 1

Publications and Home Runs around Tenure



Moral hazard in healthcare

Gaining insurance: price of healthcare is lower $\rightarrow$ individuals consume more healthcare

$\Longrightarrow$ A rational response to an economic incentive?

$\Longrightarrow$ What is the price elasticity of demand for *healthcare*?

History of the term "moral hazard"

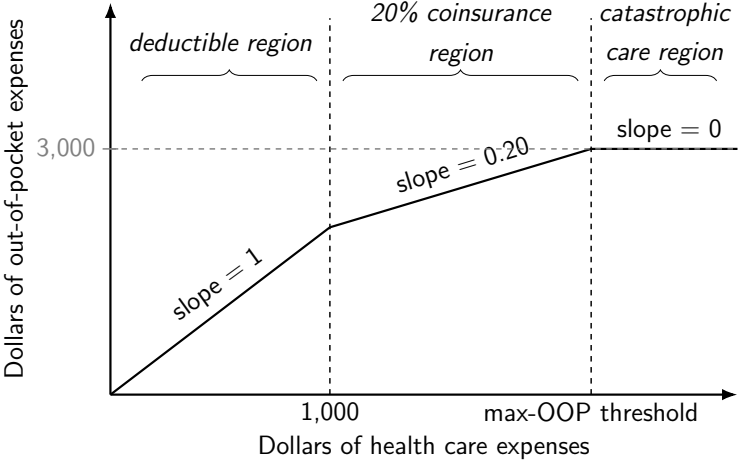
Kenneth Arrow (1963):*"the special economic problems of medical care can be explained by adaptations to uncertainty"*

Mark Pauly (1968): *"in addition to the issues of risk and uncertainty, insurers also have to anticipate how consumers will respond to the changes in the price of healthcare caused by insurance"*

- Explains that the term "moral hazard" comes from medical insurance writers describing "the hazard that arises from the failure of individuals who are or have been affected by insurance to uphold the accepted moral quantities"...
- ... but Pauly notes that "the response of seeking more medical care with insurance ... is a result not of moral perfidy, but of rational economic behavior"

Reminder from last week

Cost-sharing: portion of medical bill that the patient is responsible for paying



How to think about it?

Insurer chooses:

1. **Premium**: p
2. **Coinsurance rate**: s

Imagine the individual is sick and needs to pay $25,000 in medical bills with probability 0.2.

$\implies$ What is the insurer's problem?

$$\max_{p,s} p - 0.2 * (1 - s) * 25,000$$

such that

$$EU(\text{insured}) \geq EU(\text{not insured})$$

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Moral hazard: $d > 0$

$\implies$ The individual would consume $\$25,000 \times (1 + d \times s)$ in medical care if insured.

$\implies$ The insurer faces a trade-off:

- Higher $s \rightarrow$ less moral hazard $\rightarrow$ lower costs
- Higher $s \rightarrow$ less attractive insurance $\rightarrow$ lower premium

$\implies$ Is $d = 0$? If not, how large is it?

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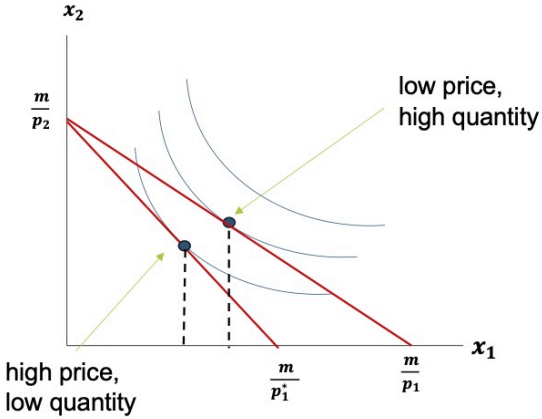
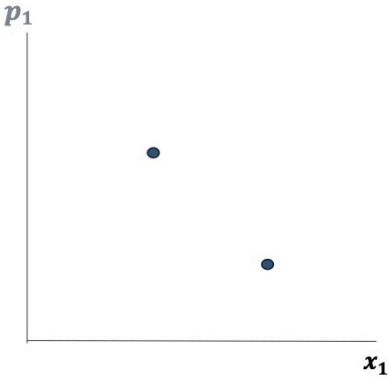
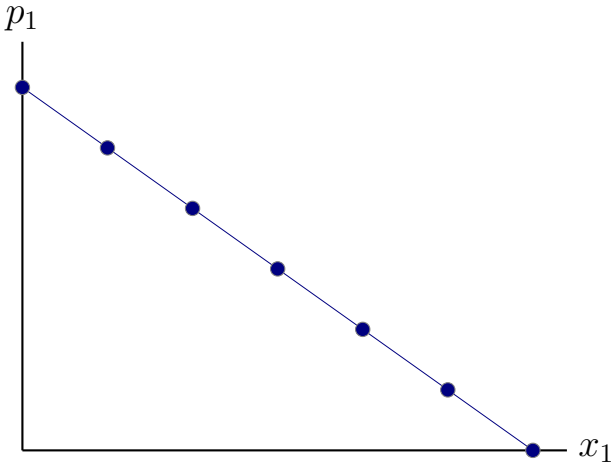
Back to basics of demand

Demand

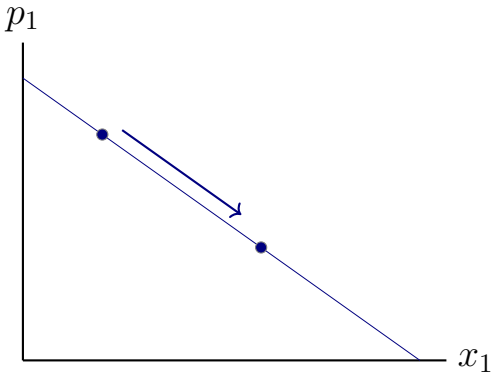

x_2
$\frac{m}{p_2}$
low price,
high quantity
high price,
low quantity
$\frac{m}{p_1^*}$
$\frac{m}{p_1}$
x_1


p_1
x_1

Demand



Move *along* the demand



Demand elasticity

Demand elasticity ϵ (price elasticity of demand):

$$\epsilon = \frac{\overbrace{dQ/Q}^{\text{\% change in Q}}}{\underbrace{dP/P}_{\text{\% change in P}}} = \frac{dQ}{dP} \times \frac{P}{Q}$$

Example: Assume the demand is $P = 200 - 25Q$.

1. What is the expression for the elasticity?
2. What is the elasticity at $P = 100$?
3. Use the elasticity to predict the change in demand from going to $P = 100$ to $P = 25$.

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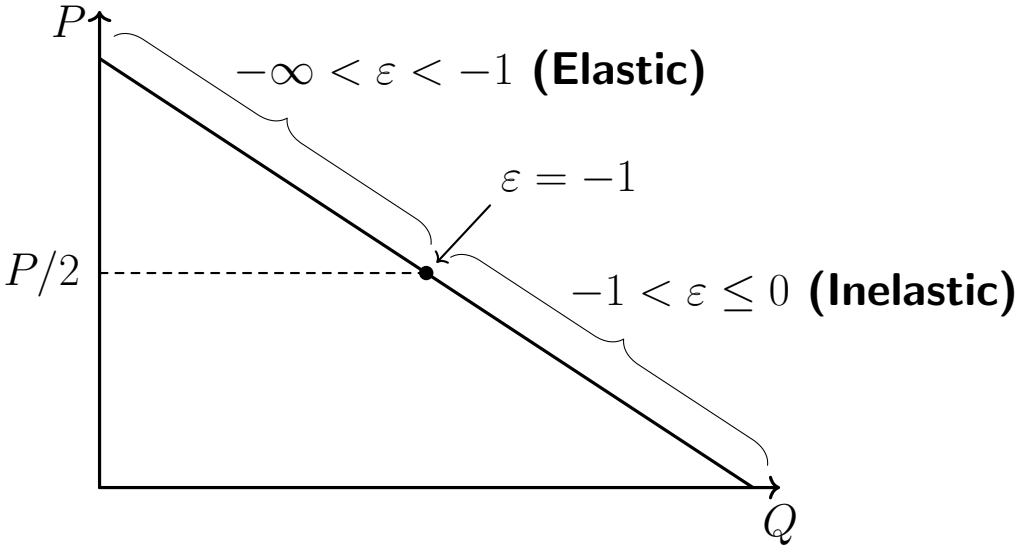
Example: Assume the demand is $P = 200 - 25Q$.

1. What is the expression for the elasticity? $\epsilon = -\frac{P}{25Q}$
2. What is the elasticity at $P = 100$? $\epsilon = -1$
3. Use the elasticity to predict the change in demand from going to $P = 100$ to $P = 25$. At $P = 100$, $Q = 4$ and $\epsilon = -1$. We have $-1 = \frac{dQ}{dP}\frac{P}{Q} = \frac{dQ}{-(100-25)}\frac{100}{4}$.
   Solving for dQ gives $dQ = 3$ and $Q' = 4 + 3 = 7$.

Demand elasticity



Demand elasticity: some examples

Table: Price Elasticity of Demand

Good	$\lvert\varepsilon\rvert$
Airline travel	2.4
Restaurant meals	2.3
Housing	1.2
Movies	0.9
Gasoline (LR)	0.7
Coffee	0.25
Automobiles	0.2
Gasoline (SR)	0.2
Outpatient care	0.17
Inpatient care	0.14

Is the demand for healthcare different than other goods?

What is cost-sharing?

Cost-sharing: portion of medical bill that the patient is responsible for paying

Demand slopes **down**

1. Is this true in healthcare?
2. If so, to what extent?
3. What does this mean for insurance?

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The **RAND health insurance experiment**

Back to the late 70s in California...

Reading: the RAND health insurance experiment

The Health Insurance Experiment

A Classic RAND Study Speaks to the Current Health Care Reform Debate

After decades of evolution and experiment, the U.S. health care system has yet to solve a fundamental challenge: delivering quality health care to all Americans at an affordable price. In the coming years, new solutions will be explored and older ideas revisited. One idea that has returned to prominence is cost sharing, which involves shifting a greater share of health care expense and responsibility onto consumers. Recent public discussion of cost sharing has often cited a landmark RAND study: the Health Insurance Experiment (HIE). Although it was completed over two decades ago, in 1982, the HIE remains the only long-term, experimental study of cost sharing and its effect on service use, quality of care, and health. The purpose of this research brief is to summarize the HIE's main findings and clarify its relevance for today's debate.

Key findings:

- In a large-scale, multiyear experiment, participants who paid for a share of their health care used fewer health services than a comparison group given free care.
- Cost sharing reduced the use of both highly effective and less effective services in roughly equal proportions. Cost sharing did not significantly affect the quality of care received by participants.
- Cost sharing in general had no adverse effects on participant health, but there were exceptions: free care led to improvements in hypertension, dental health, vision, and selected serious symptoms. These improvements were concentrated among the sickest and poorest patients.

The **RAND health insurance experiment**

The **RAND health insurance experiment**

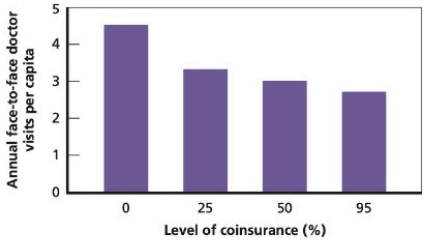
- Joe Newhouse (PI) at the RAND corporation
- 1974-1981: randomize $\approx$ 2,000 families into plans with **different cost-sharing**
- Duration of experiment: 3-5 years
- 6 enrollment sites
- Outcomes of interest: health spending and health

A pioneering study

- One of the largest experiments ($300 million in 2011 dollars)
- Screening questionnaires
- The experiment **acts as your insurer**

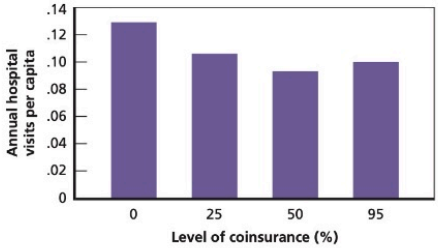
The **RAND health insurance experiment**

Figure 1
Participants with Cost Sharing Visited the Doctor Less Frequently



SOURCE: Newhouse and the Insurance Experiment Group, 1993, Tables 3.2 and 3.3.
NOTE: Utilization numbers include both adults and children.

Figure 2
. . . and Were Admitted to Hospitals Less Often



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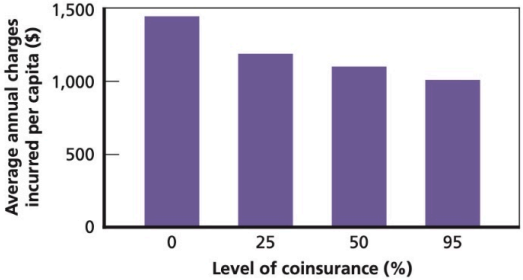
The **RAND health insurance experiment**

Figure 3
Participants with Cost Sharing Spent Less on Health Care Services



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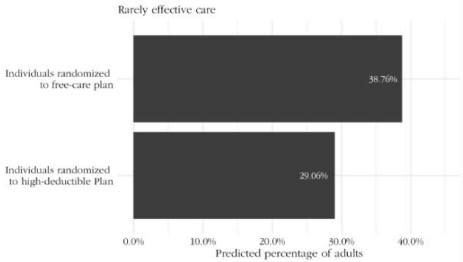
NOTES: Spending numbers include both adults and children. Spending numbers have been adjusted to 2005 dollars using all-items Consumer Price Index.

Which kind of care is consumed less?



This figure shows the percentage reduction in total healthcare consumed for "highly effective care" and "rarely effective care". These categories are defined by physicians based on their expert judgement about the average effectiveness of different types of care. For example, receiving medical care for some types of acute conditions (like bronchitis, vaginitis, or lacerations) is often highly effective; by contrast, receiving medical care for other types of acute conditions (e.g., sprains or strains) is rarely effective and can be managed with over-the-counter medication. The figure shows a similar percentage reduction in both types of healthcare when comparing families randomized into the high-deductible plans and the free-care plans. Simply put, deductibles cause families to cut back on both highly effective care and rarely effective care.

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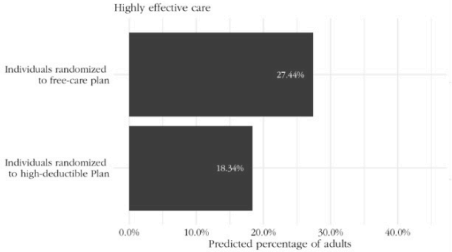

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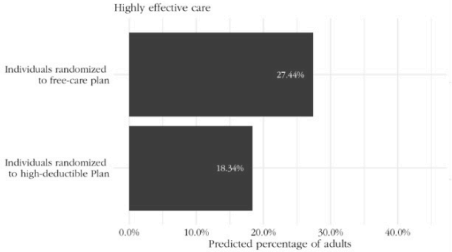

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$\Longrightarrow$ Which care is cut? Both "rarely effective" and "effective" care.

Impact on health?

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Effects on Health

In general, the reduction in services induced by cost sharing had no adverse effect on participants' health. However, there were exceptions. The poorest and sickest 6 percent of the sample at the start of the experiment had better outcomes under the free plan for 4 of the 30 conditions measured. Specifically,

- Free care improved the control of *hypertension*. The poorest patients in the free care group who entered the experiment with hypertension saw greater reductions in blood pressure than did their counterparts with cost sharing. The projected effect was about a 10 percent reduction in mortality for those with hypertension.
- Free care marginally improved *vision* for the poorest patients.
- Free care also increased the likelihood among the poorest patients of receiving needed *dental care*.
- *Serious symptoms*[2] were less prevalent for poorer people on the free plan.
- Cost sharing also had some beneficial effects. Participants in cost sharing plans worried less about their health and had fewer restricted-activity days (including time spent in seeking medical care).

The **RAND health insurance experiment**: conclusion

Main results:

- Healthcare demand slopes down (elasticity $\approx$ -0.2)
- Insurance increases the demand for healthcare
- Which care is cut: both "effective" and "rarely effective" care
- Impact on health? None on average but poor adults are hurt

Legacy

- Largest and most expensive RCT in social science to date
- No impact on health: paved the way for cost-sharing (80s) and high-deductible plans (2000s)

What is the impact of health insurance?

Insurance reduces the prices faced by patients
$\rightarrow$ increases the demand for healthcare
$\rightarrow$ increases healthcare utilization

A trade-off:

- Some healthcare is not very effective for some patients $\rightarrow$ more insurance $\rightarrow$ small gains and high costs
- Some healthcare is effective $\rightarrow$ more insurance $\rightarrow$ better health $\rightarrow$ could lead to less spending!

What is the impact of health insurance?

What do we want to know?

$$y_i = \pi_0 + \pi_1 \text{INSURANCE}_i + X_i' \pi_2 + \epsilon_i$$

where

- y_i is an outcome of interest for individual i (e.g., healthcare utilization, health, financial health)
- INSURANCE_i is whether i has insurance
- X_i are observables on individual i (e.g., income, gender, age, etc.)

Parameter of interest? .

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Parameter of interest? π_1.

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Problem?

What is the impact of health insurance?

What do we want to know?

$$y_i = \pi_0 + \pi_1 \text{INSURANCE}_i + X_i' \pi_2 + \epsilon_i$$

Problem? **Endogeneity of insurance**.

This means that INSURANCE_i is correlated with ϵ_i:

$$cov(\text{INSURANCE}_i, \epsilon_i) \neq 0$$

Thus OLS estimates of π_1 will be biased and inconsistent.

What do we do? *We find some exogenous variation in insurance coverage.* How? Let's see this next week.

Wraping up

What we covered today

- Moral hazard: definitions
- Reminders on demand elasticity
- The RAND Health Insurance Experiment

To next class

- **TO DO**
  1. **Problem Set 2** due by **9:30am next Thursday 02/12**
  2. Review this class and past material $\Longrightarrow$ *questions on it?*

- Program for next class
  - The impact of health insurance
  - Other ways to reduce moral hazard